CLASS 8 — MATHS · FORMULA SHEET

Chapter 10: **Proportional Reasoning-2** | Quick revision, one page

Direct proportion

- x and y vary directly if $\frac{x}{y}$ is constant, i.e. $\frac{x_1}{y_1} = \frac{x_2}{y_2}$.
- Increase in one quantity gives a proportional increase in the other (cost and quantity, distance and time at fixed speed).

Inverse proportion

- x and y vary inversely if xy is constant, i.e. $x_1 y_1 = x_2 y_2$.
- Examples: workers and days (fixed work), speed and time (fixed distance).
- Work $=$ (number of workers) $\times$ (number of days).

Scale and maps

- Scale $= \frac{\text{map distance}}{\text{actual distance}}$; actual distance $=$ map distance $\times$ scale factor.
- Convert to the same unit: 1 km = 1,00,000 cm.

Pie charts

- Sector angle $= \frac{\text{value of category}}{\text{total}} \times 360^\circ$.
- Percentage of a category $= \frac{\text{angle}}{360^\circ} \times 100$.
- All sector angles add up to 360° and all percentages to 100%.

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